

E-Commerce Product Recommendation System

- **Problem Statement**

E-commerce websites contain a large number of products, making it difficult for users to find similar or relevant items. This project aims to develop a Product Recommendation System that suggests the top 5 similar products based on product category, brand, and description using TF-IDF Vectorization and Cosine Similarity, helping users find suitable products quickly and easily.

- **Objective**

The main objective of this project is to develop an E-Commerce Product Recommendation System that recommends similar products based on their category, brand, and description. The system helps users discover relevant products quickly and improves their online shopping experience.

- **Purpose / Client Requirements**

The purpose of this project is to help users easily find similar products on an e-commerce platform. The client requires a system that allows users to search for a product, view the top 5 recommended similar products, and apply category and price filters through a simple and user-friendly Streamlit dashboard.

- **Methodology**

The project was developed in a step-by-step process. First, the product dataset was collected and loaded using Pandas. The data was then cleaned by removing duplicates, handling missing values, and checking data types. Exploratory Data Analysis (EDA) was performed to understand the dataset. A new Features column was created by combining the Category, Brand, and Description columns. The text data was converted into numerical form using TF-IDF Vectorization, and Cosine Similarity was used to find similar products. Finally, a Streamlit dashboard was developed to allow users to search for products and view recommendations.

- **Technologies Used :**

The project was developed using Python as the programming language. Pandas was used for data manipulation, Matplotlib for data visualization, Scikit-learn for TF-IDF Vectorization and Cosine Similarity, and Streamlit for building the interactive web application.

- **Exploratory Data Analysis (EDA) :**

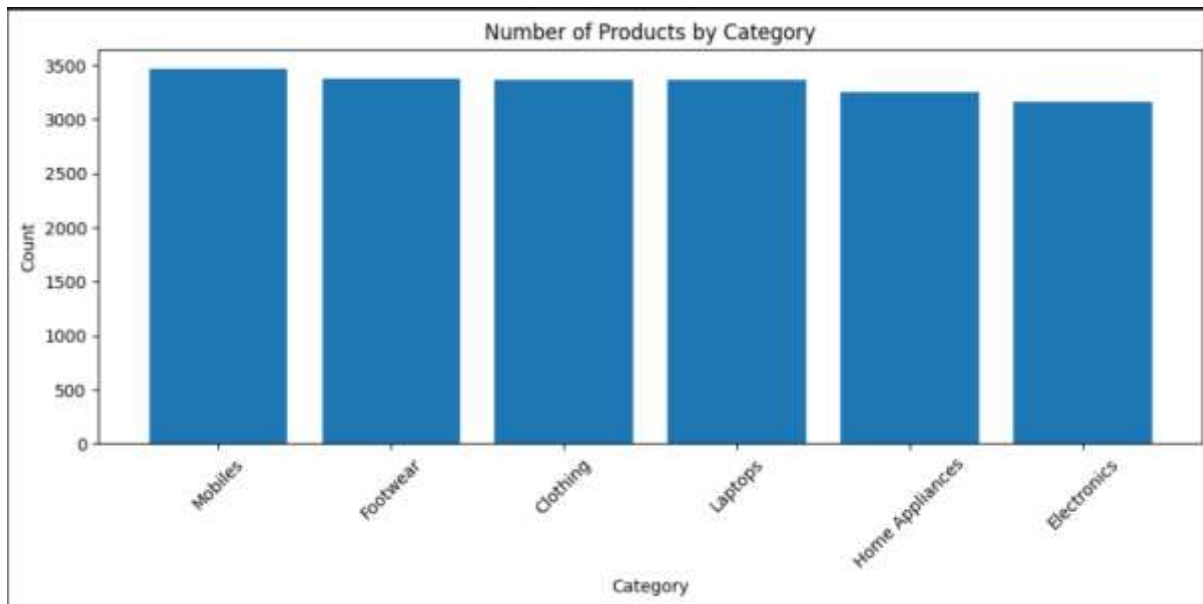
Exploratory Data Analysis (EDA) is an important step in any data science project. It helps in understanding the dataset, identifying patterns, and gaining useful insights before building the recommendation system. In this project, EDA was performed on the e-commerce product dataset to analyze the distribution of products, categories, prices, and other important attributes.

The analysis began by checking the total number of products and the number of unique product categories available in the dataset. This provided an overview of the dataset and helped in understanding its size and diversity. The top brands and the average product price were also analyzed to identify the most common brands and the overall pricing trend. In addition, the highest-rated products were identified to understand which products received the best customer ratings.

To better understand the data, different types of visualizations were created:

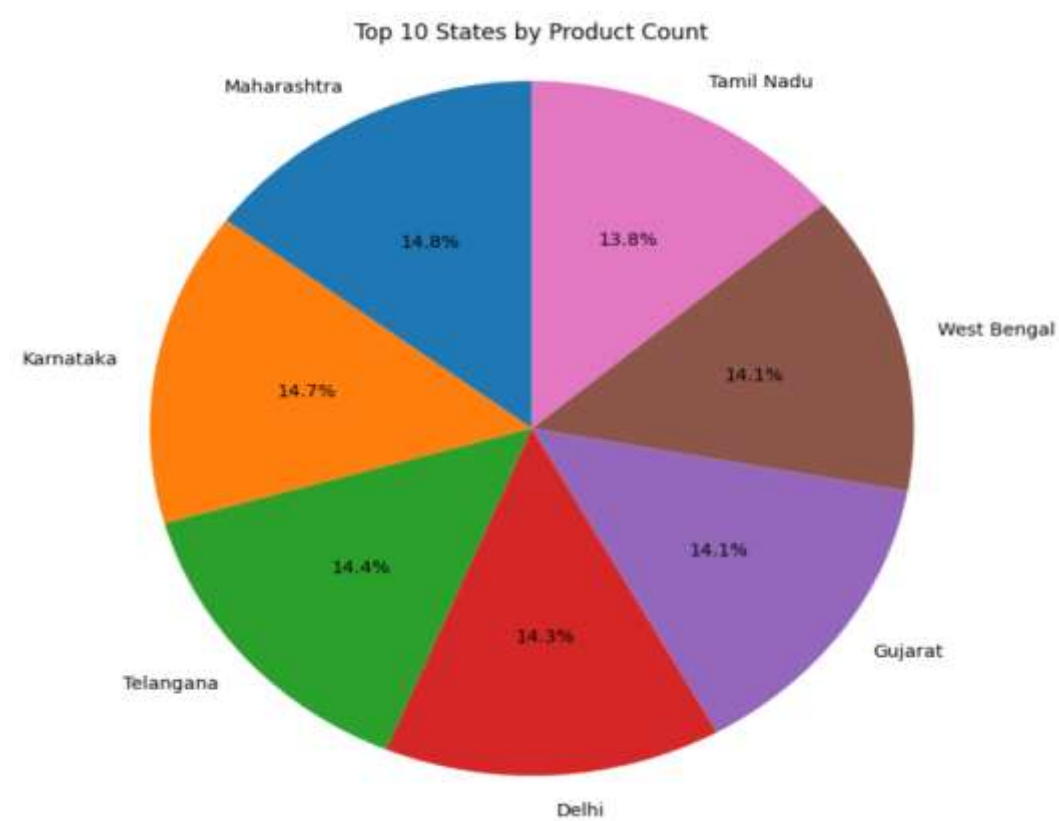
1. Bar Chart – Number of Products by Category

A bar chart was created to display the number of products available in each category. This chart helps compare the distribution of products across different categories and identifies which categories contain the highest and lowest number of products. It provides a clear overview of product availability and category-wise distribution in the dataset.



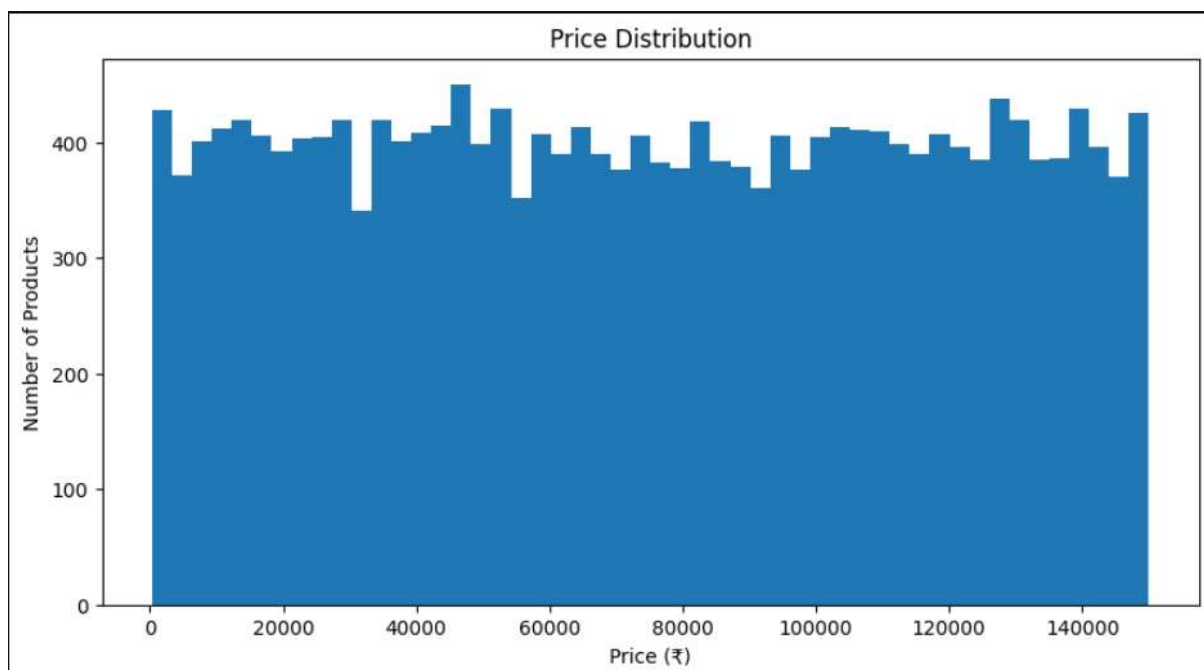
2. Pie Chart – Product Distribution by Brand

A pie chart was used to show the percentage distribution of products among different brands. This visualization helps understand the contribution of each brand to the overall dataset and makes it easy to identify the brands with the largest and smallest product collections.



3. Histogram – Distribution of Product Prices

A histogram was created to analyze how product prices are distributed across the dataset. It shows the frequency of products within different price ranges, helping to identify whether most products are low-priced, mid-range, or expensive. This visualization also helps detect any unusual price values or outliers.



Overall, the Exploratory Data Analysis provided a better understanding of the product dataset and confirmed that the data was suitable for building the recommendation system. The insights obtained from EDA also helped in the feature engineering process and supported the development of an accurate product recommendation model.

visualizations such as bar charts, pie charts, and histograms were also created to identify patterns and gain useful insights from the dataset.

- **Feature Engineering**

Feature Engineering is the process of creating new features from existing data to improve the recommendation system. In this project, a new column named Features was created by combining the Category, Brand, and Description columns. This combined text was used to compare products and generate accurate recommendations.

- **Streamlit Dashboard :**

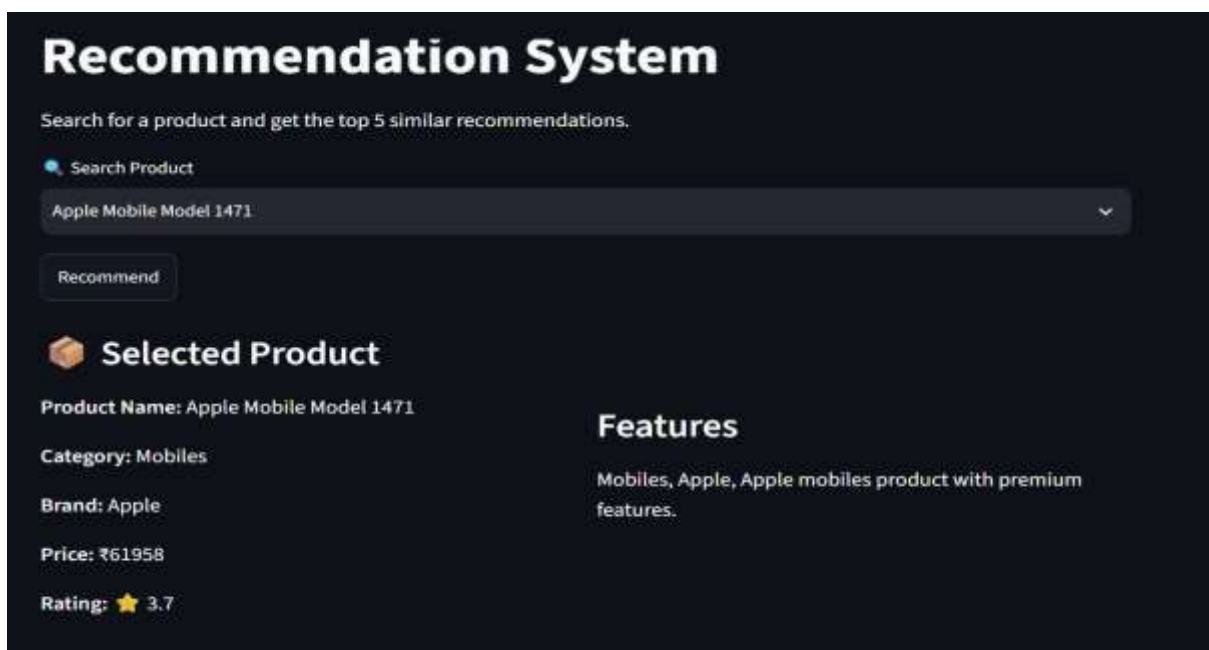
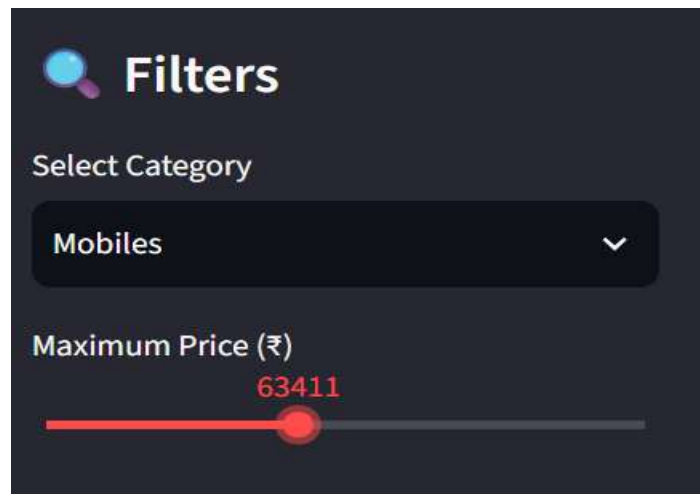
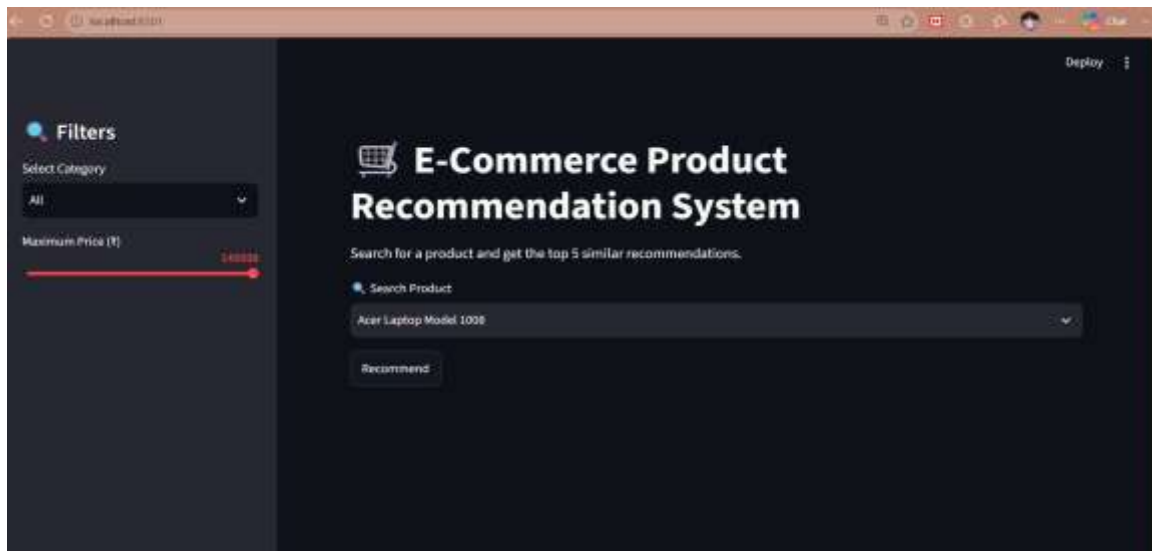
A Streamlit dashboard was developed to provide a simple, interactive, and user-friendly interface for the E-Commerce Product Recommendation System. Instead of running the program through the command line, users can easily interact with the system using a web-based application.

The dashboard includes a Product Search Box, where users can enter the name of a product they want to search for. After clicking the Recommend button, the system processes the input and displays the Top 5 similar products based on the product's category, brand, and description. For each recommended product, the dashboard displays important details such as Product Name, Category, Brand, Price, and Rating, allowing users to compare products easily.

To make the recommendations more useful, the dashboard also includes Category and Price filters. These filters allow users to narrow down recommendations according to their preferences and budget. Additionally, the application handles invalid or empty inputs by displaying appropriate error messages, ensuring a smooth user experience.

Overall, the Streamlit dashboard makes the recommendation system easy to use, interactive, and accessible, providing users with quick and relevant product recommendations through a clean and responsive interface.

Dashboard:



★ Top 5 Recommended Products

Apple Mobile Model 3899

Category: Mobiles

Brand: Apple

Price: ₹19832

Rating: ★ 4.9

Features: Mobiles, Apple, Apple mobiles product with premium features.

Apple Mobile Model 2523

Category: Mobiles

Brand: Apple

Price: ₹40964

Rating: ★ 3.6

Features: Mobiles, Apple, Apple mobiles product with premium features.

Apple Mobile Model 2160

Category: Mobiles

Brand: Apple

Price: ₹33956

• Conclusion

This project successfully developed an E-Commerce Product Recommendation System using TF-IDF Vectorization and Cosine Similarity. The system provides relevant product recommendations, helping users find similar products quickly and improving their overall shopping experience.

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